

1. (20 marks) For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = (a + b)x_1^2 + (a - b)(2x_1x_2 + x_2^2).$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a + b & a - b \\ a - b & a - b \end{pmatrix}$$

(2 marks) Its second order principal minor is equal to $2b(a - b)$. Its first order principal minors are equal to $a + b$ and $a - b$.

- (b) (12 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all the values for a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*?

Solutions: (2 marks) A quadratic form is indefinite if $Q(\mathbf{x}) > 0$ and $Q(\mathbf{y}) < 0$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$.

(4 marks) $Q(\mathbf{x})$ is positive semidefinite if all principal minors are positive which requires:

$$a \geq b \geq 0.$$

The condition accounts for scenarios in which some of the parameters are equal to 0.

(4 marks) $Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

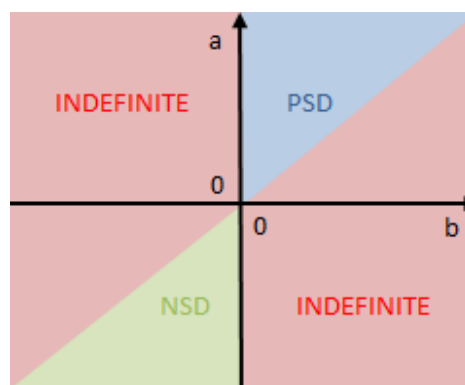
$$a \leq b \leq 0.$$

The condition accounts for scenarios in which some of the parameters are equal to 0.

(2 marks) $Q(\mathbf{x})$ is indefinite whenever it is not semidefinite.

- (c) (4 marks) Plot the pairs (a, b) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$.

Solutions: (4 marks) The plot below depicts the answer



2. (20 marks) Consider the functions $f(\mathbf{x}) = x_1^2 e^{-x_2}$ defined on some convex set U in $\mathbb{R}_+ \times \mathbb{R}$.

- (a) (6 marks) Define what it means for f to be *quasi-concave* and *quasi-convex*.

Solutions: The function f is: (3 marks) *quasi-concave* on U if $C_k^+ = \{\mathbf{x} \in U \mid f(\mathbf{x}) \geq k\}$ is a convex set for every real number k ; (3 marks) *quasi-convex* on U if $C_k^- = \{\mathbf{x} \in U \mid f(\mathbf{x}) \leq k\}$ is a convex set for every real number k .

- (b) (14 marks) Is f *quasi-concave* or *quasi-convex*? Hint: Use counterexamples to show that a property fails.

Solutions: (9 marks) The function f is quasi-concave. To show this, define on the same domain U , the function

$$g(\mathbf{x}) = 2 \log x_1 - x_2.$$

Note that g must be quasi-concave on U consider because for any $k \in \mathbb{R}$, any $\mathbf{x}, \mathbf{y} \in C_k^+$, and any $a \in [0, 1]$, we have that

$$\begin{aligned} g(a\mathbf{x} + (1-a)\mathbf{y}) &= 2 \log(ax_1 + (1-a)y_1) - (ax_2 + (1-a)y_2) = \\ &\geq a2 \log(x_1) + (1-a)2 \log(y_1) - (ax_2 + (1-a)y_2) \\ &= ag(\mathbf{x}) + (1-a)g(\mathbf{y}) \geq k, \end{aligned}$$

where the first inequality follows by the concavity of the logarithm and the second inequality follows from $\mathbf{x}, \mathbf{y} \in C_k^+$. But if g is quasi-concave, so is f since it is an increasing transformation of g ,

$$f(\mathbf{x}) = e^{g(\mathbf{x})}.$$

- (c) (5 marks) The function f is not quasi-convex. To show this, consider $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (e, 2)$. Note that $f(\mathbf{x}) = f(\mathbf{y}) = 1$, meaning that $\mathbf{x}, \mathbf{y} \in C_1^-$. Let $\mathbf{z} = \mathbf{x}/2 + \mathbf{y}/2$. If so, we have that $\mathbf{z} \notin C_1^-$ since

$$f(\mathbf{z}) = (e/2 + 1/2)^2 e^{-1} = (e + 2 + e^{-1})/4 > 1.$$

3. (40 marks) Consider the problem:

$$\max_{x,y} e^{xy} \quad \text{s.t.} \quad x + y \leq a, \quad x \geq 0, \quad ay \geq 1.$$

- (a) (10 marks) For what values of $a > 0$ is Slater condition satisfied? For such values of a , do Kuhn-Tucker necessary first order conditions identify a solution to the constrained optimization problem?

Solutions: (5 marks) To check whether Slater condition holds, we need to find values of $a > 0$ such that the interior of the constrained set is non-empty. This is the case for all $a > 1$, since the point $\tilde{x} = \varepsilon/2$ and $\tilde{y} = 1/a + \varepsilon/2$ always belongs to the interior of the constraint set for ε sufficiently small as long as $a > 1$, given that for $\varepsilon \in (0, a - 1/a)$, we have that

$$\tilde{x} + \tilde{y} = 1/a + \varepsilon < a, \quad \tilde{x} > 0, \quad \tilde{y} > 1/a.$$

Slater condition fails for $a = 1$, since a single point belongs to the constraint set, and for $a < 1$ as the constraint set is empty.

(5 marks) For $a > 1$, Kuhn-Tucker necessary conditions identify a solution to the constrained optimization problem, since all constraints are linear and thus quasi-convex, and because the objective function is quasi-concave, as there is an increasing transformation of the objective such that the transformed objective is concave. To show this consider,

$$g(x, y) = \log(\log(f(x, y))) = \log x + \log y.$$

As g is a sum of concave functions, it is quasi-concave, and thus f must be quasi-concave too.

- (b) (6 marks) Write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem.

Solutions: (6 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= ye^{xy} - \lambda_1 + \lambda_2, \\ 0 &= xe^{xy} - \lambda_1 + a\lambda_3, \\ 0 &= \lambda_1(x + y - a) = \lambda_2x = \lambda_3(1 - ay), \\ 0 &\geq x + y - a, \quad 0 \geq -x, \quad 0 \geq 1 - ay, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

- (c) (18 marks) Solve the constrained optimization problem for all values of a satisfying the restrictions found in part (a).

Solutions: Consider the various supports:

(3 marks) At the solution, $\lambda_2(a) = 0$. Otherwise, $f(x(a), y(a)) = 1$ which cannot be optimal, since the point $\tilde{x} = \varepsilon/2$ and $\tilde{y} = 1/a + \varepsilon/2$ always belongs to the constraint set for ε sufficiently small and

$$f(\tilde{x}, \tilde{y}) > 0.$$

(3 marks) At the solution, $x(a) + y(a) = a$. To show this, posit that $x(a) + y(a) < a$, and consider any hypothetical solution $(x(a), y(a))$ and the point $\tilde{x} = x(a) > 0$ and $\tilde{y} = a - x(a) > y(a)$. Such point always belongs to the constraint set and

$$f(\tilde{x}, \tilde{y}) > f(x(a), y(a)),$$

leading to a contradiction.

(6 marks) If $\lambda_2(a) = 0$ and $\lambda_1(a), \lambda_3(a) > 0$, then by complementary slackness $x(a) = a - 1/a$ and $y(a) = 1/a$ for $a > 1$. Further, FOC would then imply that

$$\begin{aligned} \lambda_1(a) &= (1/a)e^{1-1/a^2} > 0 \\ \lambda_3(a) &= (2/a^2 - 1)e^{1-1/a^2} \geq 0 \Leftrightarrow 2/a^2 \geq 1. \end{aligned}$$

So as multipliers are non-negative, this is a solution for $a \leq \sqrt{2}$.

(6 marks) Finally, if $\lambda_3(a) = \lambda_2(a) = 0$ and $\lambda_1(a) > 0$, then FOC and the constraints would imply that

$$\begin{aligned} y(a) &= x(a) = a/2 \geq 0, \\ \lambda_1(a) &= (a/2)e^{a^2/4} \geq 0, \\ ay(a) &= a^2/2 \geq 1 \Leftrightarrow a \geq \sqrt{2}. \end{aligned}$$

So as constraints need to be satisfied, this is a solution for $a \geq \sqrt{2}$.

- (d) (6 marks) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to express the slope of value function with respect to the parameter a in terms of the solution $(x(a), y(a), \boldsymbol{\lambda}(a))$ found in part (c).

Solutions: (6 marks) The Envelope Theorem states that if $(x(a), y(a), \boldsymbol{\lambda}(a))$ are differentiable functions and the constraint qualification holds, then

$$\frac{dv(a)}{da} = \frac{\partial L(x, y, \boldsymbol{\lambda}|a)}{\partial a} \Big|_{(x, y, \boldsymbol{\lambda})=(x(a), y(a), \boldsymbol{\lambda}(a))} = \lambda_1(a) + \lambda_3(a)y(a) > 0.$$

4. (20 Marks) Consider the correspondence $\mathbf{h} : [0, 4]^2 \rightrightarrows [0, 4]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} [0, 3] & \text{if } x_2 \in [0, 2] \\ [1, 4] & \text{if } x_2 \in [2, 4] \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [0, 2] & \text{if } x_1 \in [1, 4] \\ [2, 4] & \text{if } x_1 \in [0, 3] \end{cases}.$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence.

Solutions: (4 marks) Both correspondences have closed graph and are therefore UHC. Moreover, both are convex-valued and non-empty. Since the image of $\mathbf{h}$ belongs to $[0, 4]^2$ which is also the domain, the Fixed Point Theorem applies. Thus, there will be at least one fixed point on $[0, 4]^2$.

(5 marks) First, consider any point $\bar{\mathbf{x}} \in [1, 3] \times [0, 4]$. Such a point is a fixed point of $\mathbf{h}$ since

$$h_2(\bar{\mathbf{x}}) = [0, 4] \quad \text{and} \quad [1, 3] \subset h_1(\bar{\mathbf{x}}).$$

(5 marks) Next, consider any point $\bar{\mathbf{x}} \in [0, 4] \times \{2\}$. Such a point is a fixed point of $\mathbf{h}$ since

$$h_1(\bar{\mathbf{x}}) = [0, 4] \quad \text{and} \quad 2 \in h_2(\bar{\mathbf{x}}).$$

(3 mark) Further, there can be no fixed point when $x_1 < 1$ and $x_2 \neq 2$, since for any such $\mathbf{x}$

$$h_2(\mathbf{x}) = [2, 4] \quad \text{but} \quad h_1(\mathbf{x}) = [1, 4] \quad \text{for any } x_2 > 2.$$

(3 marks) Finally, there can be no fixed point when $x_1 > 3$ and $x_2 \neq 2$, since for any such $\mathbf{x}$

$$h_2(\mathbf{x}) = [0, 2] \quad \text{but} \quad h_1(\mathbf{x}) = [0, 3] \quad \text{for any } x_2 < 2.$$